

1. In what ways is denial of service (lack of availability for authorized users) a vulnerability to users of single-user personal computers?

Answer: For a single user PC, a denial of service attack can make the system or key applications unavailable to its rightful user. This could happen through malware consuming all CPU or memory resources, a ransomware lockout, or an operating system crash caused by malicious code. Even though it is not a shared system, the lack of access to files, internet, or system functions prevents the user from performing normal work. And this effectively denies service to the authorized owner.

2. Identify the three most probable threats to a computing system in an office with fewer than ten employees. That is, identify the three vulnerabilities most likely to be exploited. Estimate the number of times each vulnerability is exploited per year. Justify your estimate.

Answer: The three most probable threats in a small office (fewer than ten employees) are:

1. Phishing or email scams (about 5/7 times a year): Employees may receive fraudulent emails that steal credentials or install malware which is very common and often successful due to limited technical knowledge and wisdom.
2. Malware infections from downloads or peripherals (3/4 times a year): Without strict device policies, an infected attachment or drive can spread ransomware or spyware easily.
3. Accidental data loss or hardware failure (2/3 times a year): small office often lack automated backups, so a crashed hard drive or deleted file can result in lost information.

3. When is an incident over? That is, what factors influence whether to continue the work of the incident handling team or to disband it?

Answer: An incident is considered over when the immediate threat has been contained, systems are restored to normal operation, and no evidence of continuing compromise remains. The decision to end incident response depends on factors such as full system recovery, confirmation that vulnerabilities are patched, logs show no ongoing suspicious activity, and all affected data or users have been addressed.